

Aidan Holton

Software Engineer | Data & Applied ML | CS/ML Student

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EDUCATION

University of North Georgia, B.S. in Computer Science, Minor in Machine Learning

Aug 2022 - May 2027

Relevant Coursework: Software Engineering, Machine Learning, AI in Manufacturing, Web Programming, Databases, Introduction to UNIX, Secure Software Development, Digital Information Processing, Data Structures and Algorithms, Networking & Communications

TECHNICAL SKILLS

Java, JavaScript, TypeScript, Python, SQL, HTML, CSS, Next.js, React.js, Node.js, Express.js, Spring Boot, LangChain, YOLO, Pytesseract, Docker, AWS, Redis, Supabase, PostgreSQL, SQLite, Git, GitHub, SSMS, Microsoft Report Builder, CVAT, Ollama

WORK EXPERIENCE

University of North Georgia, Strategic Impact & Engagement

Dahlonega, GA

Data Engineer Intern | SSMS, SQL, Microsoft Report Builder, Lucidchart

Aug 2026 - Current

- Automated UNG's Common Data Set (CDS) reporting by building analytical SQL tables in SQL Server (SSMS) that generate form answers directly from source data and surfacing them through Microsoft Report Builder, replacing a manual compilation process that had required a dedicated role costing roughly \$40-50k per year.
- Owned the project end to end, spanning data discovery, analytical table design, and query development.
- Partnered with Institutional Research to locate and interpret data, running meetings with IR staff to map CDS requirements.

Platinum TCG

Cumming, GA

Software Developer | Next.js, AWS S3, AWS SES, Redis, Supabase, Vercel, Claude Code

Aug 2026 - Current

- platinum-tcg.com Built a Next.js storefront and admin console for a trading card shop using the App Router, React Server Components for data loading and Server Actions for every form, with Postgres on Supabase behind it.
- Integrated AWS S3 for images with short-lived presigned URLs for anything a customer uploaded, AWS SES for transactional email, and Redis counters for rate limiting the public forms by IP.

Gilat DataPath (via Georgia Tech PIN Program)

Duluth, GA

Software Engineer Intern | Python, YOLO, Pytesseract, PDFPlumber, CVAT, Docker, Claude Code

May 2026 - July 2026

- Rebuilt and significantly improved a computer vision pipeline that validates hardware engineering drawings against their Bill of Materials, cutting manual review time from ~10 minutes to 1-3 minutes per drawing.
- Trained a YOLO object detection model to 99.5% accuracy for detecting part-number balloons on drawings, using SAHI tiling for small-object detection and a self-annotated dataset built in CVAT.
- Engineered OCR and table-extraction stages using Tesseract and pdfplumber, handling structural edge cases including split tables, CID-encoded fonts, and fallback extraction paths.
- Built a driver generation tool that parses vendor SNMP MIB files and produces TCL device drivers automatically, replacing manual config writing with a two-command workflow; integrated an optional local LLM for gap-filling ambiguous fields. Currently in use and confirmed to increase productivity for the software team.

University of North Georgia, Strategic Impact & Engagement

Dahlonega, GA

Data Engineer Intern | SSMS, SQL, TSQL, JSON, Lucidchart

Jan 2026 - May 2026

- Developed structured JSON semantic definitions for reporting views as part of a RAG AI prototype, establishing a consistent semantic format to support future view integration within the semantic layer.
- Wrote SQL queries to extract and aggregate institutional data, building relational tables for UNG's annual Common Data Set to support standardized reporting requirements.
- Designed and documented logical ER diagrams for each domain within UNG's Enterprise Data Warehouse, improving visibility into schema structure and standardizing domain-level documentation across the warehouse.

PROJECTS

Lucrum | TypeScript, Next.js, Node.js, Express.js, Docker, AWS, Supabase, LangChain, Git

Feb 2026 - Apr 2026

- Designed and deployed a full-stack, AI-powered personal finance application using a microservices architecture with four independently deployed services; frontend live at lucrumproject.com.
- Built a Next.js frontend with secure JWT authentication via HTTP-only cookies and middleware-based route protection, connected to three separate backend REST APIs.
- Integrated Claude (via LangChain's ChatAnthropic) into a dedicated Account Summary Service to generate personalized, AI-powered financial summaries and actionable recommendations from a user's income and expense data.
- Implemented separate Account Management and Income Management services in Node.js/Express, handling user authentication with bcrypt password hashing and scoped PostgreSQL queries via Supabase.
- Containerized all backend services with Docker and deployed to AWS EC2 with custom subdomains per service; backend later scaled down for cost, frontend still live on Vercel.